

ONEDRAFT

CAD-AI — Aria Powered

Global Geospatial Intelligence, Mapping & Earth Data Integration Architecture

Version 1.0

Document: OneDraft Global Geospatial Intelligence, Mapping & Earth Data Integration Specification

System: OneDraft CAD-AI

Intelligence Layer: Aria

Document Class: Architectural & Implementation Stack Specification

Status: Architectural Expansion Specification

Predecessor: OneDraft End-to-End System Integration Specification

Successor: OneDraft Global Spatial Data Implementation & Planetary Repository Specification

1. PURPOSE

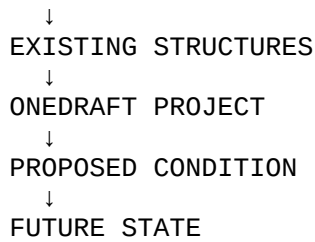
The OneDraft Global Geospatial Intelligence, Mapping & Earth Data Integration Specification establishes the architectural framework through which OneDraft extends beyond individual architectural projects into a persistent, geographically indexed representation of the built environment and associated Earth-system information.

Stack 51.0 introduces the global geospatial architecture required to connect OneDraft projects to geographic location, existing structures, sites, cities, regions, countries, continents and eventually a planetary-scale spatial information system.

The architecture establishes the foundation for integrating OneDraft with geographic information systems, including ArcGIS and other interoperable GIS platforms, while preserving OneDraft as the authoritative application and architectural information environment.

The purpose of this stack is to establish the system through which:

```
WORLD
↓
CONTINENT
↓
COUNTRY
↓
REGION
↓
CITY
↓
SITE
↓
EXISTING CONDITIONS
```



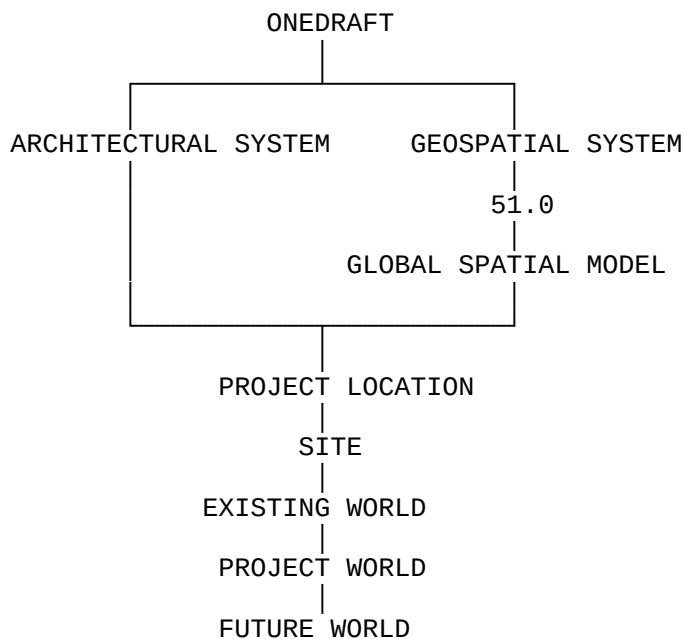
can become part of one continuously extensible geospatial information architecture.

2. SYSTEM EXTENSION

Stacks 1.0–50.0 establish OneDraft as an integrated CAD-AI architectural project platform.

Stack 51.0 introduces the **Global Geospatial Layer**.

The resulting architecture becomes:



The geospatial architecture therefore becomes a system-level extension of the existing OneDraft platform rather than an independent mapping application.

3. GLOBAL GEOSPATIAL OBJECTIVE

The long-term objective of Stack 51.0 is to establish a geographically indexed OneDraft repository capable of associating architectural information with physical locations throughout the world.

The system shall support progressive geographic expansion:

PROJECT
↓
SITE
↓
CITY
↓
REGION
↓
COUNTRY
↓
CONTINENT
↓
GLOBAL

The architecture shall permit the repository to grow incrementally rather than requiring planetary-scale data to exist before the system becomes operational.

A OneDraft project may therefore contribute a geographically meaningful record without requiring the entire surrounding geographic environment to already exist within the OneDraft repository.

4. GEOSPATIAL SYSTEM PRINCIPLE

The fundamental principle of Stack 51.0 is:

Every OneDraft project may be geographically situated within a controlled spatial reference framework, while every external geographic dataset incorporated into OneDraft retains its provenance, source, temporal state, transformation history and applicable access restrictions.

The system shall distinguish between:

Geographic Location

Spatial Geometry

Architectural Geometry

External GIS Data

OneDraft Project Data

Derived Data

Historical Data

Current Data

Proposed Data

Future Data

These categories shall not be treated as interchangeable.

5. GLOBAL SPATIAL MODEL

The Global Spatial Model shall provide the common geographic framework through which OneDraft projects and external spatial datasets can be associated.

The model shall support:

Coordinates

Geographic Boundaries

Sites

Parcels where legally and technically available

Buildings

Structures

Infrastructure

Transportation

Land Use

Topography

Water Bodies

Environmental Features

Administrative Boundaries

Project Locations

Spatial Relationships

The model shall remain extensible to future Earth-system datasets.

6. SPATIAL HIERARCHY

The global repository shall support hierarchical geographic relationships.

PLANET



CONTINENT



COUNTRY



ADMINISTRATIVE REGION



MUNICIPALITY / CITY
↓
DISTRICT
↓
SITE
↓
PROPERTY / GEOGRAPHIC UNIT
↓
STRUCTURE
↓
ARCHITECTURAL ELEMENT

The hierarchy shall support multiple geographic classification systems rather than assuming that all jurisdictions use the same administrative structure.

7. PROJECT GEOREFERENCING

Every OneDraft project shall be capable of receiving a geographic identity.

Project georeferencing may include:

Latitude

Longitude

Elevation

Coordinate Reference System

Site Boundary

Building Footprint

Orientation

Local Coordinate System

Survey Reference

Geographic Identifier

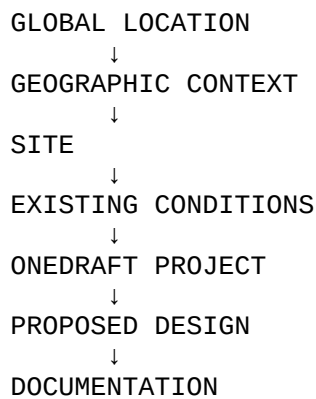
Jurisdiction

The project geographic identity shall become part of the project model.

8. PROJECT-TO-WORLD RELATIONSHIP

The OneDraft project shall no longer exist solely as an isolated architectural object.

Its relationship shall become:



This relationship allows architectural information to remain associated with its real-world context.

9. EXISTING STRUCTURE REGISTRY

Stack 51.0 shall establish an architectural framework for registering pre-existing structures where appropriate data is available and legally usable.

Existing structures may include:

Residential Buildings

Commercial Buildings

Institutional Buildings

Industrial Buildings

Infrastructure

Public Facilities

Transportation Structures

Historic Structures

Other Built Assets

The registry shall distinguish between known, inferred, imported and OneDraft-generated information.

10. EXISTING CONDITIONS

The Global Spatial Model shall support existing-condition records.

Existing-condition information may originate from:

GIS

CAD

BIM

Survey

Public Geographic Data

Licensed Data

Client Data

Site Documentation

Remote Sensing

Imagery

Other Authorized Sources

Imported information shall retain source provenance.

11. ARCGIS INTEGRATION

OneDraft shall establish an interoperability layer for ArcGIS and related geographic information services.

ArcGIS shall be treated as an integrated geospatial platform rather than as the authoritative definition of the OneDraft architecture.

The integration layer shall support appropriate ArcGIS resources and services, including where applicable:

Feature Services

Feature Layers

Map Services

Geospatial Data

Spatial Queries

Geographic Features

Layer Metadata

Geographic Attributes

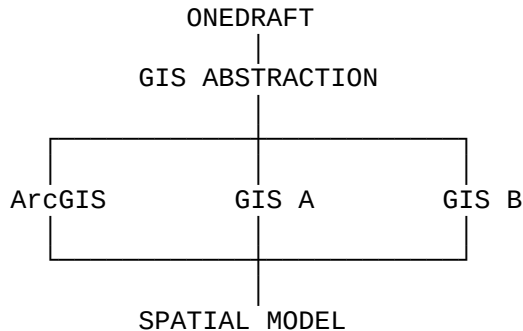
Spatial References

Integration shall occur through controlled adapters.

12. GIS PROVIDER ABSTRACTION

OneDraft shall not permanently bind the Global Spatial Model to a single GIS provider.

The architecture shall provide a GIS abstraction layer:

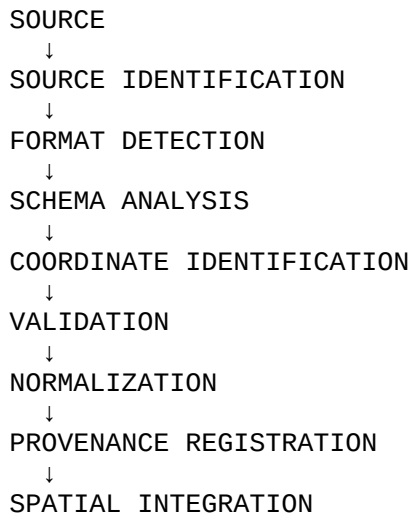


This permits future integration with additional GIS providers and geographic data systems.

13. SPATIAL DATA INGESTION

External spatial information shall enter OneDraft through controlled ingestion pipelines.

The ingestion process shall include:



Invalid or incompatible information shall not silently enter authoritative project state.

14. COORDINATE REFERENCE SYSTEM

The geospatial architecture shall explicitly represent coordinate reference systems.

Spatial records shall preserve:

Coordinate Reference System

Projection

Datum

Units

Transformation

Precision

Source Reference

Where transformation occurs, the system shall preserve the transformation metadata necessary to reproduce or understand the conversion.

15. SPATIAL NORMALIZATION

Different external datasets may use different:

Coordinate Systems

Units

Schemas

Identifiers

Geometry Types

Levels of Precision

Temporal References

Administrative Definitions

OneDraft shall normalize these differences before incorporating the data into the common spatial model.

Normalization shall not destroy the original source representation.

16. SPATIAL PROVENANCE

Every significant external spatial object shall maintain provenance.

Provenance may include:

Source

Provider

Dataset

Layer

Record Identifier

Acquisition Date

Effective Date

Coordinate System

Transformation

Processing History

Version

License

Access Classification

The system shall distinguish source information from information derived by OneDraft.

17. TEMPORAL GEOSPATIAL MODEL

The Global Spatial Model shall support time.

A structure shall not be represented solely as a static geographic object.

The system shall support:

HISTORICAL

↓

EXISTING

↓

CURRENT

↓

ALTERED

↓

PROPOSED

↓

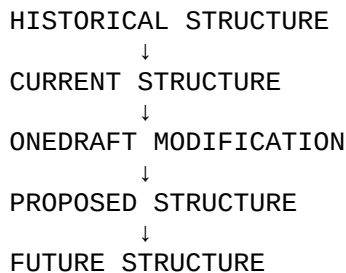
FUTURE

Historical states shall remain distinguishable from current and proposed states.

18. WORLD-STATE RECONCILIATION

The system shall provide mechanisms for reconciling information representing different moments in time.

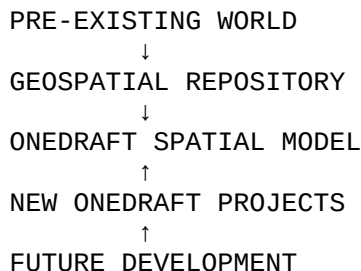
For example:



The system shall preserve the temporal lineage rather than overwriting previous states without controlled versioning.

19. EXISTING + NEW CONVERGENCE

The central architectural concept of Stack 51.0 is the convergence of existing-world information and newly generated OneDraft information.



Over time, historical and newly generated information can occupy the same spatial and temporal framework.

This establishes the architectural basis for a continuously evolving built-environment repository.

20. SPATIAL PROJECT REGISTRY

Each OneDraft project shall have a spatial registry record.

The registry shall associate:

Project ID

Organization

Site

Geographic Coordinates

Spatial Boundary

Jurisdiction

Existing Structures

Project Model

Project Status

Project Version

Temporal State

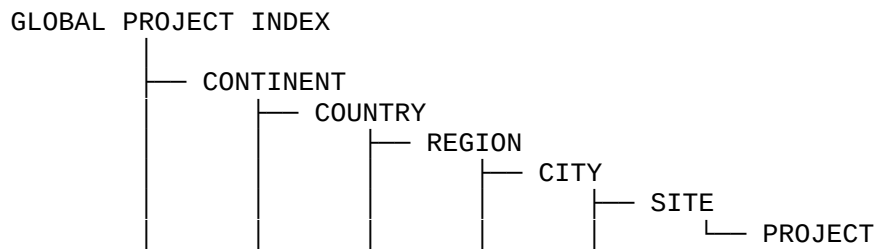
Artifacts

The registry shall permit projects to be discovered through authorized geographic queries.

21. GLOBAL PROJECT INDEX

OneDraft shall support a global project index.

The conceptual structure shall be:



The index shall support both geographic and project-centric navigation.

22. SPATIAL SEARCH

The system shall support spatial discovery.

Potential query types include:

Projects within an area

Structures near a project

Projects within a city

Projects within a jurisdiction

Projects intersecting a boundary

Projects within a radius

Structures associated with a site

Historical states of a location

Spatial searches shall respect authorization and data-access controls.

23. SPATIAL RELATIONSHIPS

The Global Spatial Model shall support relationships such as:

Contains

Contained By

Intersects

Adjacent To

Within

Overlaps

Near

Connected To

Derived From

Replaces

Modified From

Located At

These relationships shall permit geographic reasoning across datasets.

24. BUILT-ENVIRONMENT MODEL

The OneDraft global repository shall progressively represent the built environment.

The model may encompass:

SITE

↓
BUILDING
↓
LEVEL
↓
SPACE
↓
ASSEMBLY
↓
ELEMENT

while simultaneously connecting that architectural hierarchy to:

CITY
↓
REGION
↓
COUNTRY
↓
CONTINENT
↓
PLANET

The result is a multi-scale representation of architectural and geographic information.

25. ARCHITECTURAL-GEOGRAPHIC INTEGRATION

Architectural geometry and geographic geometry shall remain related but distinct.

GEOGRAPHIC REFERENCE
↓
SITE COORDINATE SYSTEM
↓
BUILDING POSITION
↓
LOCAL PROJECT COORDINATE SYSTEM
↓
ARCHITECTURAL GEOMETRY

This permits detailed architectural modeling without sacrificing geographic positioning.

26. CAD / GIS CONVERGENCE

CAD and GIS data shall be interoperable through controlled transformation.

The architecture shall support:

CAD Geometry → GIS Geometry

GIS Geometry → Project Context

Survey → Project Model

GIS Boundary → Site Context

OneDraft Geometry → Geographic Representation

Transformation shall preserve precision and provenance.

27. BIM / GIS CONVERGENCE

Where BIM information is available, OneDraft shall support relationships between:

BIM Objects

Building Geometry

Geographic Location

Site Context

Infrastructure

GIS Features

The system shall avoid treating BIM and GIS as identical representations.

They shall instead form complementary views of the same physical environment.

28. GLOBAL SPATIAL REPOSITORY

The Global Spatial Repository shall provide persistent storage and indexing for authorized geospatial information.

Repository classes may include:

Projects

Sites

Structures

Boundaries

Infrastructure

Environmental Features

Spatial Metadata

Historical States

Derived Features

External Dataset References

The repository shall support large-scale spatial indexing and retrieval.

29. SPATIAL DATABASE ARCHITECTURE

The database architecture shall support spatial data types and spatial indexing.

The implementation shall accommodate:

Points

Lines

Polygons

Multipolygons

Geometry Collections

Bounding Boxes

Spatial Relationships

Coordinate Systems

Temporal Attributes

Spatial persistence shall integrate with the existing OneDraft database architecture.

30. SPATIAL INDEXING

Spatial indexing shall support efficient geographic queries.

The architecture shall support indexing strategies appropriate for:

Project Locations

Building Footprints

Sites

Administrative Boundaries

Infrastructure

Large Geographic Datasets

Indexing shall be designed for progressive scaling from individual projects to regional and global datasets.

31. PLANETARY SCALE

The architecture shall be capable of eventual expansion from local project data to planetary-scale spatial indexing.

The progression shall be:

ONE PROJECT
↓
MANY PROJECTS
↓
CITY
↓
REGION
↓
COUNTRY
↓
CONTINENT
↓
GLOBAL

The system shall not require planetary-scale infrastructure at initial deployment.

It shall instead establish architectural boundaries capable of supporting progressive expansion.

32. DATA QUALITY

Global spatial information will vary in quality.

The system shall therefore represent:

Accuracy

Precision

Completeness

Currency

Source Reliability

Resolution

Confidence

Processing State

Data quality shall be represented independently from the existence of the record itself.

33. DATA CONFIDENCE

Spatial records may have different confidence classifications.

The architecture may distinguish:

Verified

Source Confirmed

Imported

Derived

Estimated

Unresolved

Superseded

Confidence shall not be silently interpreted as legal or factual certainty beyond the underlying source.

34. SPATIAL DATA LINEAGE

The system shall preserve the transformation path:

```
SOURCE DATA
↓
INGESTION
↓
NORMALIZATION
↓
TRANSFORMATION
↓
SPATIAL MODEL
↓
DERIVED FEATURE
↓
PROJECT CONTEXT
```

Every derived spatial representation shall remain traceable to its source where technically and legally appropriate.

35. TEMPORAL VERSIONING

Spatial information shall support versioning.

A structure may therefore have:

VERSION 1
↓
VERSION 2
↓
VERSION 3
↓
CURRENT
↓
PROPOSED

Previous versions shall remain identifiable according to retention and governance rules.

36. CHANGE DETECTION

The architecture shall provide a future integration boundary for identifying changes in geographic conditions.

Potential changes include:

New Construction

Alteration

Expansion

Removal

Demolition

Replacement

Land-Use Change

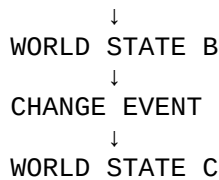
Infrastructure Change

Changes shall be represented as temporal events or state transitions where appropriate.

37. GLOBAL CHANGE MODEL

The eventual global repository shall support the conceptual relationship:

WORLD STATE A
↓
CHANGE EVENT



This transforms the repository from a static map into a temporal information system.

38. ENVIRONMENTAL DATA ARCHITECTURE

Stack 51.0 shall establish an extensible interface for future Earth-system information.

Potential future data domains include:

Land Cover

Land Use

Water

Terrain

Vegetation

Ecological Conditions

Climate Variables

Environmental Monitoring

Natural Hazards

Infrastructure Conditions

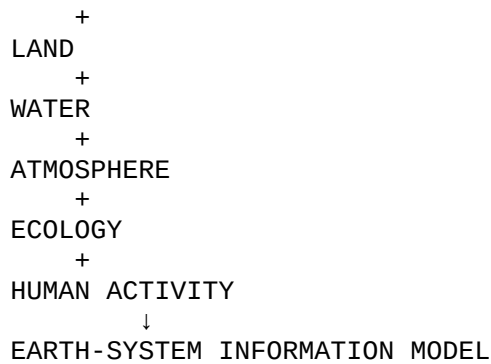
This specification establishes the interface architecture rather than asserting that all such datasets are immediately available.

39. EARTH-SYSTEM INTEGRATION

The geospatial layer shall provide the geographic foundation for future Earth-health and Earth-ecology systems.

Conceptually:

GEOGRAPHY
+
BUILT ENVIRONMENT



OneDraft shall therefore maintain a clean separation between architectural data and future environmental datasets while allowing them to become spatially interoperable.

40. HUMANITARIAN DATA INTERFACE

The architecture shall provide a future integration boundary for appropriately governed humanitarian datasets.

Potential data classes include:

Population Movement

Displacement

Shelter Capacity

Infrastructure Availability

Affected Areas

Humanitarian Facilities

Transportation Access

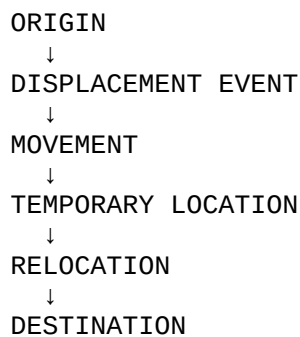
Resource Distribution

Such data shall be governed according to applicable privacy, security, licensing and humanitarian-data requirements.

41. MASS-DISPLACEMENT MAPPING

The global spatial architecture shall be capable of supporting future geographic representations of population displacement.

Conceptually:



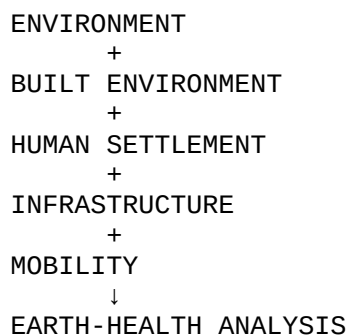
The system shall distinguish between aggregate geographic information and personally identifiable information.

The architectural objective is spatial analysis and system interoperability, not unrestricted exposure of individual persons.

42. EARTH-HEALTH DATA FOUNDATION

The spatial architecture shall provide a geographic substrate for future Earth-health analysis.

Potential relationships may include:



This capability shall remain a future extension of the geospatial architecture.

43. ARIA GEOSPATIAL INTELLIGENCE

Aria shall be capable of interacting with the geospatial system through controlled commands and queries.

Potential capabilities include:

Spatial Query Interpretation

Project Location Analysis

Geographic Context Construction

Site Context Summarization

Spatial Relationship Analysis

Dataset Discovery

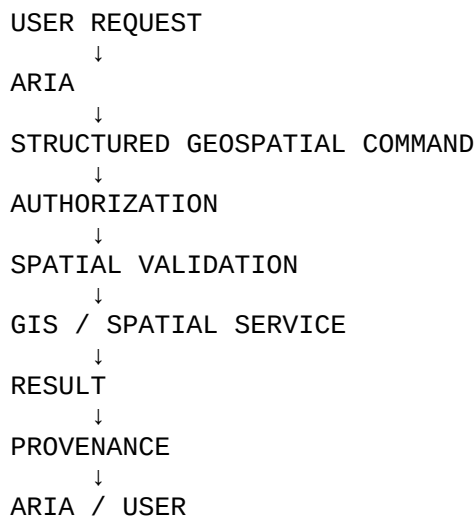
Change Analysis

Geospatial Workflow Orchestration

Aria shall not bypass spatial authorization or provenance controls.

44. GEOSPATIAL COMMAND ARCHITECTURE

Geospatial AI operations shall follow:



AI-generated spatial interpretations shall remain distinguishable from authoritative geographic source records.

45. GEOSPATIAL ARTIFACTS

Spatial operations may generate:

Maps

Layers

Spatial Reports

GeoJSON

CAD Exports

GIS Packages

Spatial Tables

Rendered Images

Project Context Packages

Analytical Outputs

Artifacts shall enter the existing OneDraft Artifact Management architecture.

46. GLOBAL MAP INTERFACE

The OneDraft frontend shall eventually support a geographic interface.

The conceptual interface shall permit:

WORLD
↓
CONTINENT
↓
COUNTRY
↓
CITY
↓
SITE
↓
PROJECT
↓
BUILDING
↓
DOCUMENTATION

The geographic interface shall remain subject to authorization, data availability and source licensing.

47. SPATIAL PROJECT DISCOVERY

Authorized users may be able to discover OneDraft projects through geographic context.

Examples include:

Projects within a city

Projects within a region

Projects within an organization-defined area

Projects associated with a site

Historical project states

Existing structures associated with a project

Spatial discovery shall respect tenant isolation and access-control policies.

48. MULTI-TENANT GEOSPATIAL SECURITY

Global spatial information introduces additional security considerations.

The architecture shall distinguish:

Public Data

Licensed Data

Organization Data

Private Project Data

Restricted Data

Sensitive Data

Administrative Data

A global spatial index shall not automatically imply global visibility of every underlying project.

49. PRIVACY & SENSITIVITY

The system shall incorporate privacy and sensitivity controls into geospatial architecture from inception.

Controls may include:

Access Restrictions

Spatial Generalization

Data Masking

Tenant Boundaries

Role-Based Access

Purpose-Based Access

Dataset Restrictions

Retention Rules

Audit Logging

The system shall avoid unnecessary exposure of sensitive location information.

50. LICENSING & DATA RIGHTS

External geographic datasets shall maintain applicable licensing metadata.

The system shall distinguish:

Owned Data

Licensed Data

Publicly Available Data

User-Provided Data

Derived Data

Restricted Data

OneDraft shall not treat access to a dataset as equivalent to ownership of the underlying dataset.

51. GEOSPATIAL AUDITABILITY

Spatial operations shall be auditable.

The audit model shall support:

WHO

↓

ACCESSED WHAT

↓

FROM WHICH SOURCE

↓

WHEN

↓

UNDER WHICH AUTHORIZATION

↓

WHAT TRANSFORMATION OCCURRED

↓

WHAT RESULT WAS GENERATED

This shall integrate with the existing Identity, Security, Provenance and Observability architecture.

52. SPATIAL EVENT BUS

Geospatial operations shall integrate with the existing OneDraft Event Bus.

Events may include:

Project Georeferenced

Spatial Dataset Imported

Spatial Object Created

Spatial Object Updated

Spatial Object Superseded

Spatial Relationship Changed

GIS Synchronization Completed

Spatial Validation Completed

Events shall remain versioned and traceable.

53. GIS SYNCHRONIZATION

The GIS Integration Layer shall support controlled synchronization.

Synchronization shall distinguish:

EXTERNAL SOURCE
↓
IMPORT
↓
NORMALIZATION
↓
ONEDRAFT RECORD

and:

ONEDRAFT PROJECT
↓
EXPORT
↓
TRANSFORMATION
↓
EXTERNAL GIS

Synchronization shall not assume that every external source permits bidirectional modification.

54. CONFLICT RESOLUTION

Where multiple sources provide conflicting geographic information, the system shall preserve the conflict rather than silently choosing one source without a defined rule.

Conflict handling may consider:

Source Authority

Recency

Accuracy

Resolution

Dataset Version

Project Evidence

Administrative Context

The resulting decision shall be traceable.

55. SPATIAL VALIDATION

Spatial data shall pass deterministic checks appropriate to its function.

Validation may include:

Coordinate Validity

Geometry Validity

Topology

Projection Consistency

Self-Intersection

Boundary Consistency

Required Attributes

Spatial Reference

Data Completeness

Invalid geometry shall not silently propagate into authoritative project records.

56. SPATIAL TOPOLOGY

The architecture shall support topology-aware relationships.

Examples include:

Adjacent Structures

Overlapping Boundaries

Contained Features

Intersecting Infrastructure

Connected Networks

Shared Boundaries

Topology shall support geographic reasoning without replacing the underlying source geometry.

57. GLOBAL REPOSITORY SCALING

The Global Spatial Repository shall be architected for progressive expansion.

The system shall support:

10 PROJECTS
↓
100 PROJECTS
↓
10,000 PROJECTS
↓
1,000,000+ RECORDS
↓
REGIONAL DATASETS
↓
GLOBAL DATASETS

Actual capacity shall be established through implementation testing rather than assumed from architectural diagrams.

58. PERFORMANCE

Spatial workloads shall be separated from ordinary transactional workloads where necessary.

Performance architecture may include:

Spatial Indexes

Caching

Tile-Based Rendering

Asynchronous Processing

Worker Pools

Partitioning

Geospatial Query Optimization

Dataset Preprocessing

Artifact Caching

Large geographic datasets shall not be permitted to degrade ordinary project operations unnecessarily.

59. JOB EXECUTION

Geospatial workloads shall integrate with the existing Job & Worker Execution System.

Jobs may include:

Dataset Import

Coordinate Transformation

Spatial Validation

Geometry Conversion

GIS Synchronization

Spatial Indexing

Map Generation

Change Detection

Large-Area Analysis

Job state shall remain persistent and observable.

60. OBSERVABILITY

The geospatial system shall integrate with OneDraft observability.

Telemetry shall include:

Import Duration

Query Latency

Dataset Size

Transformation Failures

Spatial Validation Failures

GIS Availability

Worker Utilization

Spatial Query Performance

Synchronization State

Geospatial failures shall be distinguishable from ordinary application failures.

61. BACKUP & RECOVERY

Critical spatial records shall participate in OneDraft backup and recovery.

Recovery shall account for:

Spatial Database State

Dataset Metadata

Project Spatial Relationships

Spatial Artifacts

Configuration

Integration Credentials

Processing State

External source data shall not be represented as locally owned backup material merely because metadata referencing it is retained.

62. DATA RETENTION

Spatial records shall follow controlled lifecycle policies.

The architecture shall support:

Active

Historical

Archived

Superseded

Deleted

Externally Referenced

Retention policies shall distinguish between source records, derived records and project records.

63. GEOSPATIAL CONFIGURATION

The Administrative Control Plane shall govern appropriate geospatial settings.

Configuration may include:

Supported Coordinate Systems

GIS Providers

Dataset Sources

Import Policies

Transformation Rules

Spatial Accuracy Requirements

Data Classifications

Retention Rules

Access Policies

Administrative changes shall remain versioned and auditable.

64. GEOSPATIAL TESTING

Testing shall cover:

Coordinate Transformations

Geometry Conversion

Spatial Queries

Topology

GIS Integration

Data Ingestion

Data Provenance

Temporal Versioning

Access Control

Large Dataset Performance

Failure Recovery

End-to-End Project Georeferencing

65. CI/CD INTEGRATION

Geospatial services and integration adapters shall participate in the existing CI/CD architecture.

The pipeline shall validate:

```
graph TD
    SOURCE --> BUILD
    BUILD --> UNIT_TEST[UNIT TEST]
    UNIT_TEST --> SPATIAL_TEST[SPATIAL TEST]
    SPATIAL_TEST --> INTEGRATION_TEST[INTEGRATION TEST]
    INTEGRATION_TEST --> SECURITY_TEST[SECURITY TEST]
    SECURITY_TEST --> PERFORMANCE_TEST[PERFORMANCE TEST]
    PERFORMANCE_TEST --> STAGING
    STAGING --> END_TO_END_VERIFICATION[END-TO-END VERIFICATION]
    END_TO_END_VERIFICATION --> RELEASE
```

Changes to spatial schemas or transformations shall receive appropriate migration and compatibility testing.

66. DEVELOPER PLATFORM

The Developer Platform shall eventually expose controlled geospatial interfaces.

Potential interfaces include:

Spatial API

Project Location API

GIS Integration API

Spatial Query API

Dataset Registration API

Spatial Event API

Python SDK

TypeScript SDK

Plugin Interfaces

All developer access shall remain subject to OneDraft authentication, authorization and data-governance policies.

67. EXTERNAL DATA EXCHANGE

The geospatial system shall extend Stack 40.0 through dedicated spatial exchange capabilities.

Supported data classes may include:

GIS

CAD

BIM

Survey

GeoJSON

Spatial Databases

Raster Data

Vector Data

Map Services

Feature Services

Other Authorized Geographic Formats

The external exchange layer shall remain distinct from the internal canonical spatial model.

68. SPATIAL API ARCHITECTURE

The API architecture shall expose controlled geographic resources.

Conceptual resources include:

/projects/{id}/location
/sites/{id}
/structures/{id}
/spatial/features
/spatial/layers
/spatial/datasets
/spatial/search
/spatial/relationships
/spatial/events

Actual endpoint definitions shall be established through the API and OpenAPI architecture.

69. SPATIAL DOMAIN MODEL

The domain model shall provide spatial entities including, where appropriate:

SpatialEntity

Site

Structure

SpatialFeature

SpatialDataset

SpatialReference

GeographicBoundary

SpatialRelationship

SpatialVersion

SpatialProvenance

SpatialEvent

These entities shall integrate with the existing OneDraft domain model.

70. GEOSPATIAL DOMAIN BOUNDARY

The Global Geospatial subsystem shall remain bounded.

It shall provide:

Spatial Context

Geographic Relationships

Geospatial Storage

GIS Integration

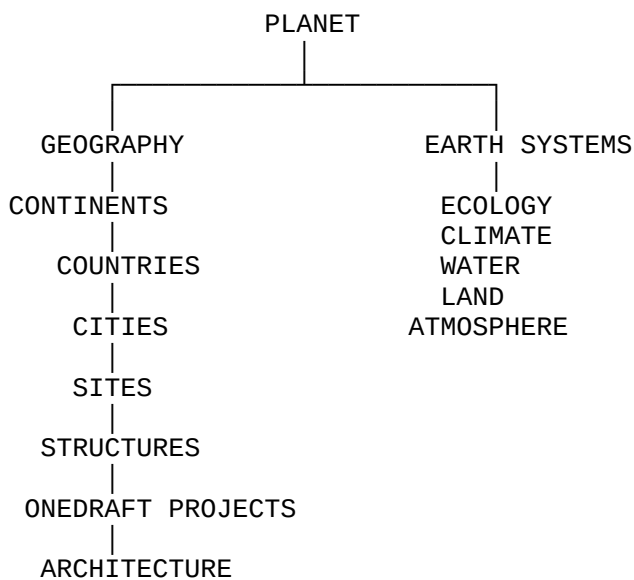
Spatial Queries

Spatial Provenance

It shall not absorb unrelated architectural domain responsibilities.

71. WORLD MODEL

The long-term conceptual model shall be:



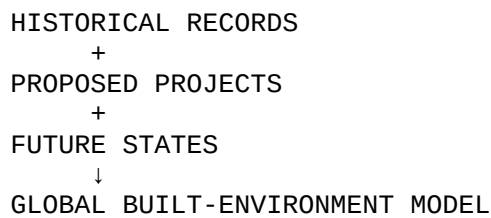
The geospatial architecture becomes the geographic connective tissue between these information domains.

72. GLOBAL BUILT-ENVIRONMENT MODEL

The accumulated OneDraft project repository may progressively form a structured representation of the built environment.

The architecture shall permit:

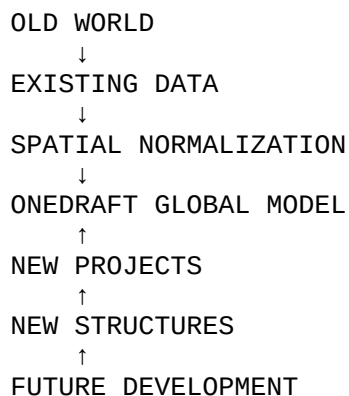
EXISTING DATA
+
ONEDRAFT PROJECTS
+



This model shall remain dependent upon available, authorized and appropriately governed data.

73. OLD + NEW CONVERGENCE

The defining long-term architectural concept of Stack 51.0 is the convergence of existing-world information and newly created OneDraft information.

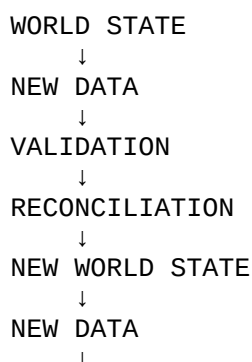


The system therefore becomes capable of maintaining a temporal bridge between the existing built environment and newly generated architectural information.

74. CONTINUOUS WORLD MODEL

The architecture shall support a continuously evolving spatial repository.

Conceptually:



...

The repository shall preserve historical states rather than treating every update as destructive replacement.

75. EARTH-ECOLOGY EXTENSION

The Global Geospatial Architecture shall provide the geographic foundation for future systems concerned with Earth ecology.

Potential future integrations may include:

Ecological Monitoring

Habitat Data

Land Change

Water Systems

Vegetation

Environmental Conditions

Natural Hazards

Climate Data

The architecture shall remain modular so that these future systems can be added without restructuring the core OneDraft project architecture.

76. EARTH-HEALTH EXTENSION

The same spatial infrastructure may eventually support broader Earth-health systems.

Conceptually:

BUILT ENVIRONMENT
+
LAND
+
WATER
+
ECOLOGY
+
CLIMATE
+
HUMAN SETTLEMENT
+
MOBILITY

Stack 51.0 establishes the spatial foundation for such future systems without defining them as part of the immediate OneDraft architectural production workflow.

77. HUMAN MOBILITY EXTENSION

The spatial model shall support future aggregate representations of human mobility and displacement where appropriately governed datasets are available.

The architecture may represent:

Origin Areas

Destination Areas

Movement Periods

Population Aggregates

Shelter Locations

Infrastructure Demand

Service Availability

The system shall maintain privacy, security and aggregation boundaries.

78. GEOSPATIAL INTELLIGENCE

Aria may eventually operate across the spatial repository to answer authorized questions such as:

WHERE?

WHAT?

WHEN?

WHAT CHANGED?

WHAT EXISTS?

WHAT IS PROPOSED?

WHAT IS CONNECTED?

WHAT IS NEARBY?

WHAT DATA SUPPORTS THIS?

Aria shall obtain answers through authorized system interfaces rather than inventing geographic facts.

79. DETERMINISTIC SPATIAL REASONING

Where spatial conclusions can be computed deterministically, OneDraft shall prefer deterministic computation.

Examples include:

Distance

Area

Intersection

Containment

Coordinate Transformation

Geometry Validity

Spatial Relationships

AI may interpret the result but shall not replace deterministic computation.

80. SPATIAL COMMAND SAFETY

Geospatial commands shall be subject to the same command-control architecture established by Stack 38.0.

```
ARIA
↓
COMMAND
↓
VALIDATION
↓
AUTHORIZATION
↓
SPATIAL SERVICE
↓
DETERMINISTIC RESULT
↓
AUDIT
```

Commands affecting persistent spatial records shall require appropriate authorization.

81. GEOSPATIAL ARTIFACT PROVENANCE

Every generated spatial artifact shall maintain:

Source Project

Source Dataset

Generation Process

Software Version

Spatial Reference

Generation Timestamp

Artifact Hash

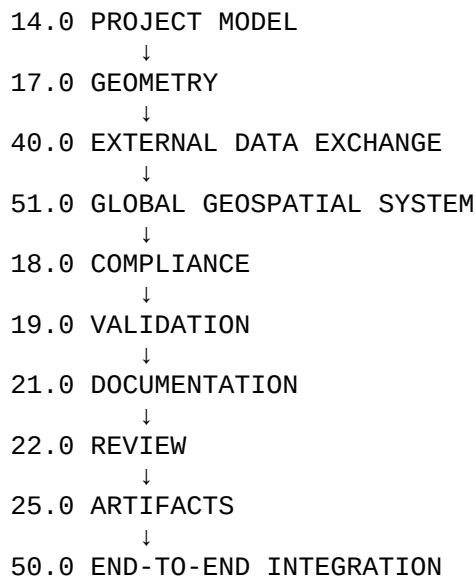
Configuration

Transformation History

This shall integrate directly with Stack 25.0 and Stack 36.0.

82. SYSTEM INTEGRATION

Stack 51.0 integrates with the existing architecture as follows:



The geospatial layer therefore becomes a cross-cutting architectural capability rather than a replacement for any existing subsystem.

83. SECURITY INTEGRATION

The geospatial architecture shall integrate with:

27.0 Identity, Access & Security

28.0 Identity, Security & Authorization

29.0 Identity, Access Control & Security

36.0 Data Lifecycle, Retention & Provenance

45.0 API Gateway, Rate Limiting & Service Protection

Security shall remain active across geographic data acquisition, processing, storage, query and export.

84. OBSERVABILITY INTEGRATION

The geospatial architecture shall integrate with Stack 30.0.

The system shall expose:

GIS Availability

Spatial Query Metrics

Dataset Processing Metrics

Spatial Job State

Transformation Failures

Data Quality Events

Synchronization Events

Access Events

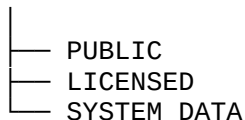
This provides system-wide operational visibility.

85. MULTI-TENANT INTEGRATION

The geospatial system shall integrate with Stack 44.0.

The architecture shall distinguish between:

GLOBAL / SHARED DATA



TENANT DATA



PROJECT DATA

Geographic indexing shall not bypass tenant boundaries.

86. BILLING INTEGRATION

Spatial capabilities may eventually be governed by Stack 43.0.

Possible metered resources may include:

Spatial Processing

Large Dataset Imports

Advanced Spatial Analysis

GIS Synchronization

Storage

Map Generation

Entitlement enforcement shall occur through existing authorization and product-control systems.

87. ADMINISTRATIVE INTEGRATION

Stack 49.0 shall govern configurable geospatial settings.

Administrative functions may include:

GIS Providers

Dataset Registries

Spatial Policies

Coordinate Systems

Import Rules

Data Classifications

Spatial Accuracy Requirements

Access Policies

Retention Rules

All administrative modifications shall remain auditable.

88. PERFORMANCE & SCALE INTEGRATION

Stack 46.0 shall govern geospatial scalability.

The architecture shall support independent scaling of:

Spatial APIs

GIS Adapters

Spatial Workers

Map Rendering

Data Ingestion

Spatial Indexing

Large-Area Analysis

This prevents geospatial workloads from becoming an architectural bottleneck for core architectural project operations.

89. DISASTER RECOVERY

Geospatial services shall participate in Stack 35.0 disaster recovery.

Recovery planning shall distinguish:

Canonical OneDraft Data

Spatial Metadata

Derived Spatial Data

External Dataset References

Cached Data

Temporary Processing Data

Not every external dataset requires local duplication.

90. OPERATIONAL READINESS

Before production deployment of major geospatial capabilities, the system shall demonstrate:

Data Integrity

Spatial Accuracy

Authorization

Provenance

Recovery

Performance

GIS Integration

Failure Handling

Observability

Security

Tenant Isolation

Operational readiness shall be demonstrated through testing rather than inferred from successful development execution.

91. GEOSPATIAL EXTENSIBILITY

The architecture shall remain extensible to future systems.

Potential future extensions include:

Digital Twin Systems

Smart-City Systems

Infrastructure Monitoring

Environmental Monitoring

Climate Systems

Disaster Response

Humanitarian Mapping

Population Movement Analysis

Ecological Systems

Earth-System Analytics

These shall integrate through controlled interfaces rather than modifying the foundational OneDraft domain model unnecessarily.

92. DATA GOVERNANCE

The Global Spatial Repository shall operate under explicit governance.

Governance shall address:

Ownership

Licensing

Privacy

Security

Retention

Provenance

Accuracy

Access

Transformation

Deletion

Archival

Governance rules shall apply according to dataset class.

93. ARCHITECTURAL INVARIANTS

The following principles are mandatory across Stack 51.0.

Spatial Authority: External GIS systems are sources and integration partners; they do not automatically become authoritative over OneDraft project state.

Provenance: Significant geographic information shall remain traceable.

Temporal Integrity: Historical states shall not be silently destroyed.

Coordinate Integrity: Spatial transformations shall be explicit.

Tenant Isolation: Private project geography shall remain private where required.

Data Governance: Geographic access shall not imply ownership.

Determinism: Spatial calculations shall use deterministic computational systems wherever practical.

AI Subordination: Aria shall not override authoritative spatial computation.

Extensibility: The GIS layer shall remain provider-neutral.

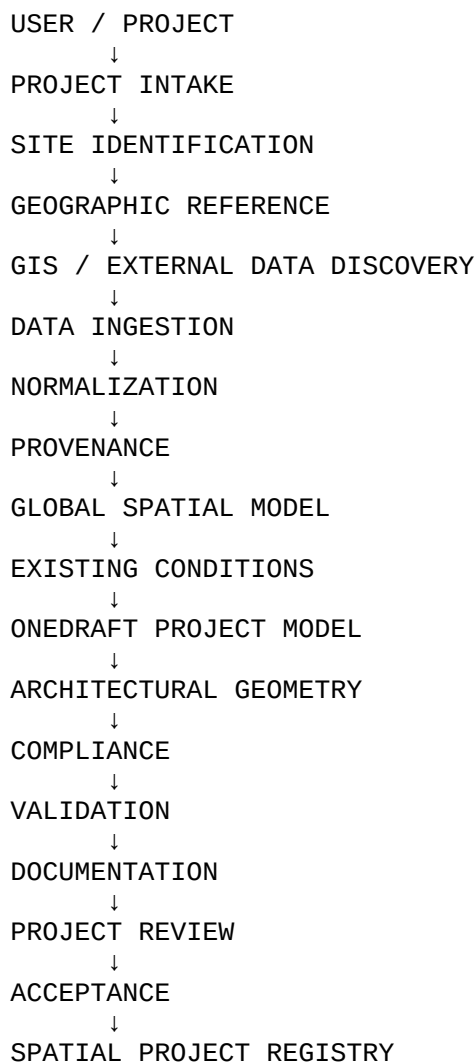
Scalability: The architecture shall permit progressive geographic expansion.

Interoperability: GIS, CAD, BIM and architectural systems shall communicate through controlled interfaces.

Auditability: Significant spatial operations shall be traceable.

94. END-TO-END GEOSPATIAL WORKFLOW

The complete workflow shall be:



↓
GLOBAL REPOSITORY

The project therefore becomes both an architectural document set and a geographically indexed system record.

95. GLOBAL DATA FLOW

The long-term global architecture shall operate:

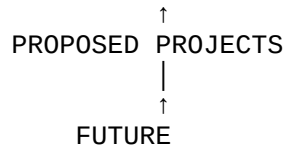
EXTERNAL WORLD
↓
GIS / SURVEY / CAD / BIM / OTHER SOURCES
↓
DATA INGESTION
↓
NORMALIZATION
↓
PROVENANCE
↓
GLOBAL SPATIAL REPOSITORY
↓
ONEDRAFT PROJECT CONTEXT
↓
ARCHITECTURAL PROJECT
↓
NEW VERIFIED DATA
↓
GLOBAL REPOSITORY

This creates a controlled feedback loop in which new OneDraft project information can become part of the broader spatial information environment where appropriate.

96. OLD WORLD / NEW WORLD MODEL

The complete temporal architecture is:

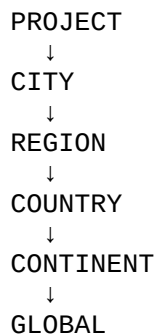
PAST
|
HISTORICAL DATA
|
↓
EXISTING WORLD
|
↓
ONEDRAFT MODEL
|
↑
CURRENT PROJECTS
|



The system therefore creates a common spatial and temporal framework in which pre-existing structures and newly developed OneDraft projects can coexist.

97. PLANETARY MAPPING TRAJECTORY

The architectural expansion shall follow:



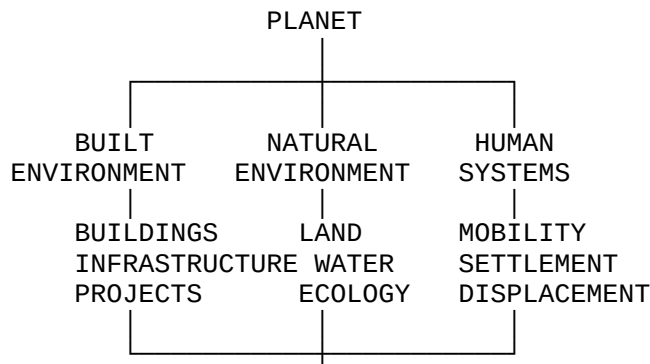
This progression shall be driven by data acquisition, authorized integration, project participation, validated geographic information and infrastructure scalability.

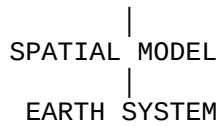
The architecture shall not require immediate global completeness.

It shall provide the structure through which completeness can progressively increase.

98. FUTURE EARTH INFORMATION SYSTEM

The long-term architecture may evolve toward:





Stack 51.0 provides the geographic architecture necessary for that future convergence.

99. RELATIONSHIP TO ONEDRAFT 1.0–50.0

Stack 51.0 does not invalidate or replace the completed 1.0–50.0 architecture.

Instead:

1.0–50.0
↓
COMPLETE ONEDRAFT CAD-AI SYSTEM
↓
51.0
GLOBAL GEOSPATIAL EXTENSION

The existing architecture remains the foundation.

Stack 51.0 becomes the first major architectural expansion beyond the original 50-stack system.

100. FUTURE AMENDMENTS TO 1.0–50.0

Following implementation of Stack 51.0, controlled amendments may be made to existing stacks where required.

Potential integration points include:

14.0 — Computational Project Model & Geometry Engine

17.0 — Geometry & Parametric Model Engine

36.0 — Data Lifecycle, Retention & Provenance System

40.0 — Import, Export & External Data Exchange

44.0 — Multi-Tenant Organization & Project Isolation

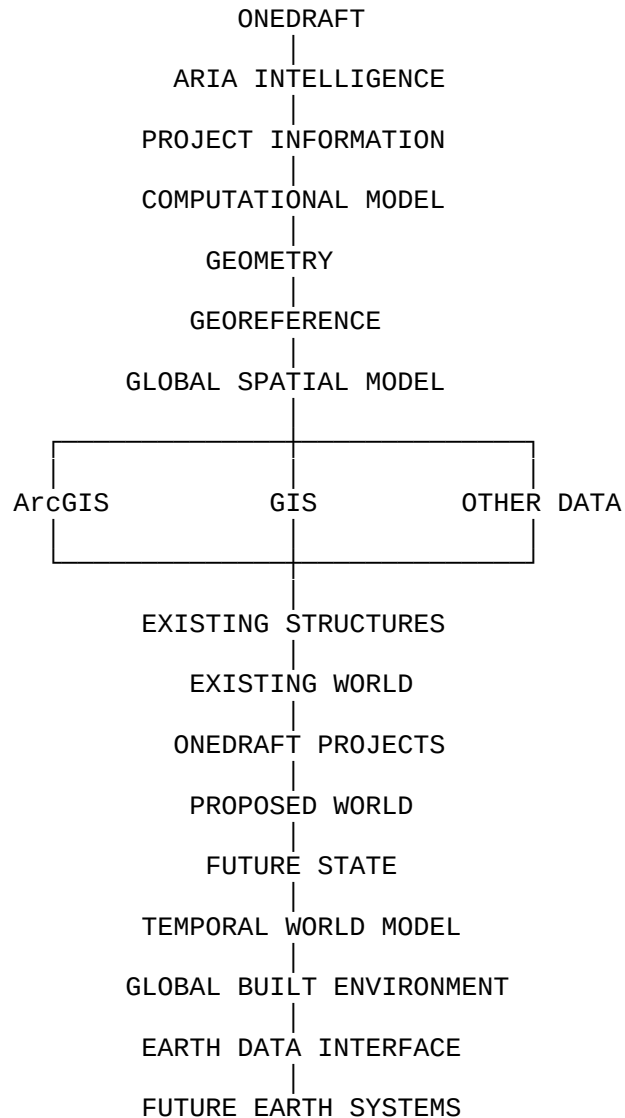
46.0 — Performance, Scalability & Capacity Architecture

50.0 — OneDraft End-to-End System Integration

These amendments shall be additive and shall preserve the architectural integrity of the completed 1.0–50.0 system.

101. MASTER GEOSPATIAL ARCHITECTURE

The complete Stack 51.0 architecture is:



102. ARCHITECTURAL COMPLETION STATE

With Stack 51.0, OneDraft acquires a new architectural dimension.

The original system is:

ARCHITECTURAL PROJECT

↓
COMPUTATIONAL MODEL
↓
DOCUMENTATION

The expanded system becomes:

WORLD
↓
GEOGRAPHIC CONTEXT
↓
SITE
↓
EXISTING CONDITIONS
↓
ARCHITECTURAL PROJECT
↓
COMPUTATIONAL MODEL
↓
DOCUMENTATION
↓
PROPOSED CONDITION
↓
GLOBAL SPATIAL REPOSITORY
↓
TEMPORAL WORLD MODEL

This transforms OneDraft from a project-isolated architectural generation platform into a system capable of maintaining geographically connected architectural information.

103. STACK 51.0 MASTER OBJECTIVE

The ultimate architectural objective of Stack 51.0 is to establish the infrastructure through which OneDraft can progressively connect:

OLD
+
CURRENT
+
NEW
+
FUTURE

within a common geographic and temporal information architecture.

The system shall permit authorized information concerning existing structures, new OneDraft projects and future project states to become spatially related without destroying source provenance or historical state.

104. FINAL INTEGRATION PRINCIPLE

The central principle of Stack 51.0 is:

EVERY PROJECT HAS A PLACE.
EVERY PLACE HAS CONTEXT.
EVERY STRUCTURE HAS A HISTORY.
EVERY CHANGE HAS A TIME.
EVERY SOURCE HAS PROVENANCE.
EVERY DATASET HAS GOVERNANCE.
EVERY FUTURE STATE CAN BE REPRESENTED.

OneDraft shall use those principles to establish a continuously extensible spatial information architecture.

105. COMPLETION STATUS

Stack 51.0 establishes the Global Geospatial Intelligence, Mapping & Earth Data Integration Architecture for OneDraft CAD-AI.

The specification establishes the architectural foundation for:

Global Project Georeferencing

ArcGIS Integration

GIS Provider Abstraction

Global Spatial Repository

Existing Structure Registration

Spatial Project Indexing

CAD/GIS Interoperability

BIM/GIS Interoperability

Spatial Data Ingestion

Spatial Normalization

Coordinate Reference Management

Spatial Provenance

Temporal Geospatial Modeling

World-State Reconciliation

Spatial Search

Spatial Relationships

Global Built-Environment Representation

Geospatial AI Operations

Spatial Security

Tenant-Aware Geographic Access

Spatial Data Governance

GIS Synchronization

Spatial Validation

Spatial Event Processing

Global Spatial Scaling

Earth-System Data Integration

Humanitarian Geospatial Data Interfaces

Future Population-Movement Analysis

Environmental and Ecological Data Integration

The resulting architectural progression is:

1.0–50.0

ONEDRAFT CAD-AI



51.0

GLOBAL GEOSPATIAL INTELLIGENCE



GLOBAL SPATIAL REPOSITORY



PROJECT → CITY → COUNTRY → CONTINENT → PLANET



HISTORICAL + EXISTING + CURRENT + PROPOSED + FUTURE



GLOBAL BUILT-ENVIRONMENT MODEL



EARTH-DATA INTEGRATION



FUTURE EARTH-SYSTEM ARCHITECTURES

Status: ARCHITECTURALLY DEFINED

Architecture Register: 1.0–51.0

System: ONEDRAFT CAD-AI

Intelligence Layer: ARIA

Geospatial Expansion: GLOBAL

Primary Geospatial Integration: GIS / ArcGIS Interoperability

Spatial Objective: GLOBAL PROJECT & BUILT-ENVIRONMENT REPOSITORY

Temporal Objective: HISTORICAL → EXISTING → CURRENT → PROPOSED → FUTURE

Long-Term Architectural Direction: GLOBAL SPATIAL & EARTH-DATA INFORMATION
SYSTEM